



SAVEETHA

INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Section-II

MCQ Questions and Answers: on: 26.02.2025

Programme : B.E / B.Tech
Course Code : CSA10
Course Name : Software Engineering

1. Rational Unified Process (RUP)

Q1: In a 12-month RUP project, if the team is 40% through the Elaboration phase (3 months long), what is the total elapsed time?

- A) 1.2 months
- B) 2.2 months
- C) 3.0 months
- D) 4.2 months

☒ B) 2.2 months

Q2: If a project is in the Construction phase and has completed 50% of the work, what percentage of the total project duration has elapsed?

- A) 50%
- B) 66.7%
- C) 75%
- D) 83.3%

☒ C) 75%

2. COCOMO Model

Q3: In the COCOMO II model, which of the following factors does *not* affect the scale factor?

- A) Precedentedness
- B) Team Cohesion

- C) Cost Drivers
- D) Risk Resolution

☒ C) Cost Drivers

Q4: If a software project has an estimated size of 150 KLOC in the Semi-Detached mode, with $E = 1.12$, what is the exponent applied to KLOC in the effort equation?

- A) 1.05
- B) 1.12
- C) 1.20
- D) 1.25

☒ B) 1.12

3. Distributed Systems & Real-time Processing

Q5: If a distributed system processes 1000 images of size 1024×1024 pixels, each pixel stored as a 16-bit integer, what is the total data size?

- A) 1 GB
- B) 2 GB
- C) 4 GB
- D) 8 GB

☒ B) 2 GB

Q6: A convolutional filter applies 9 multiplications and 8 additions per pixel. For a 1024×1024 image, how many total floating-point operations are performed?

- A) 8 million
- B) 16 million
- C) 17 million
- D) 34 million

☒ C) 17 million

4. Test-Driven Development (TDD)

Q7: If each unit test covers 20 lines of code and the codebase has 10,000 LOC, how many tests are required?

- A) 200
- B) 400
- C) 500
- D) 1000

☒ C) 500

Q8: With a unit test pass rate of 95%, how many tests are expected to fail out of 500 tests?

- A) 10
- B) 15

- C) 20
- D) 25

☒ D) 25

5. Version Control & Change Management

Q9: If 5 developers each submit one pull request per day, and each takes a full day to review before merging, what is the maximum number of parallel branches in the VCS at a time?

- A) 2
- B) 3
- C) 4
- D) 5

☒ D) 5

Q10: What is the total number of code changes merged if developers A, B, C, D, and E submit 5, 8, 10, 6, and 7 changes, respectively?

- A) 30
- B) 31
- C) 35
- D) 36

☒ D) 36

6. Financial Feasibility Analysis (NPV)

Q11: If a project has an initial cost of \$200,000, annual maintenance cost of \$25,000, and annual benefits of \$80,000 for 5 years at a 12% discount rate, what is the net annual cash flow?

- A) \$25,000
- B) \$50,000
- C) \$55,000
- D) \$60,000

☒ C) \$55,000

Q12: Which software solution should be chosen if their NPVs are:

- Solution A: -\$1,679
- Solution B: \$2,330
- Solution C: \$18,320

- A) Solution A
- B) Solution B
- C) Solution C
- D) Any of the above

☒ C) Solution C

7. Performance & Uptime Analysis

Q13: If an e-commerce platform handles 10,000 concurrent users, each making 10 transactions per minute, how many total transactions occur per hour?

- A) 60,000
- B) 600,000
- C) 6,000,000
- D) 12,000,000

☒ C) 6,000,000

Q14: A system has a 99.99% uptime requirement. How many minutes of downtime are allowed per year?

- A) 52.56 minutes
- B) 5 hours
- C) 10 minutes
- D) 100 minutes

☒ A) 52.56 minutes

8. Spiral Model & Risk Management

Q15: In a Spiral Model project with an initial cost of \$100,000, if the Engineering phase costs 50% of the total, how much is allocated for this phase?

- A) \$10,000
- B) \$20,000
- C) \$50,000
- D) \$100,000

☒ C) \$50,000

Q16: If a software project terminates after the Planning phase, and 30% of the remaining budget is lost, how much is the termination cost?

- A) \$10,000
- B) \$20,000
- C) \$27,000
- D) \$30,000

☒ C) \$27,000

9. Software Development & Defect Management

Q17: If 15 developers produce 200 LOC per day over 200 working days, how many total LOC are developed?

- A) 300,000
- B) 600,000
- C) 1,000,000
- D) 1,200,000

☒ **B) 600,000**

Q18: If only 70% of LOC are defect-free, how many defective LOC exist in a 600,000 LOC project?

- A) 100,000
- B) 120,000
- C) 180,000
- D) 200,000

☒ **C) 180,000**

10. Risk Management

Q19: A project has a 30% probability of requirement changes, with high impact (3). What is its risk exposure?

- A) 0.30
- B) 0.60
- C) 0.90
- D) 1.20

☒ **C) 0.90**

Q20: Which two risks have the highest combined risk exposure in a project?

- A) Requirement Changes & Technology Obsolescence
- B) Schedule Slippage & Security Vulnerabilities
- C) Requirement Changes & Resource Constraints
- D) Resource Constraints & Technology Obsolescence

☒ **C) Requirement Changes & Resource Constraints**

1. Rational Unified Process (RUP)

Q21: A software project following RUP is expected to last 8 months with four iterations. If the first two iterations complete 40% and 20% of the work respectively, how much work is

left for the last two iterations?

☒ C) 40%

Q22: What is the average percentage of work completed per iteration in the above case?

☒ B) 25%

2. COCOMO Model

Q23: A software project of 5000 KLOC follows the embedded mode in COCOMO II with an effort exponent of 1.11. If the effort multipliers product is 0.72, what is the estimated effort (PM)?

☒ B) ~12,850 person-months

Q24: If the team improves documentation (DOCU) from "Low" (0.90) to "Nominal" (1.00), how does it impact the total effort?

☒ A) Effort slightly increases

3. Distributed Systems & Real-time Processing

Q25: A distributed e-commerce system handles 15,000 users, each making 3 requests per second. What is the total system throughput?

☒ D) 45,000 requests per second

Q26: If the average request processing time per microservice is 40ms and there are 80 microservices, what is the effective response time per request?

☒ A) 0.5 ms

4. Test-Driven Development (TDD)

Q27: A project has 12,000 LOC and each test case covers 25 LOC. How many test cases are required for full coverage?

☒ C) 480

Q28: If the defect detection rate in TDD is 85%, how many defects might still be present after 500 tests?

☒ B) 75 defects

5. Version Control & Change Management

Q29: If 8 developers each make 12 commits per day over 40 days, what is the total number of commits?

☒ **D) 3,840 commits**

Q30: Assuming an average review time of 6 minutes per commit, how much total time is spent on code reviews?

☒ **C) 384 hours**

6. Financial Feasibility Analysis (NPV)

Q11: A software project requires an initial investment of \$300,000 and has expected annual benefits of \$120,000 for 5 years, with a 10% discount rate. What is the NPV?

☒ **B) ~\$158,000**

Q32: If a second project has an NPV of \$140,000 and similar investment costs, which project should be selected?

☒ **A) The first project**

7. Performance & Uptime Analysis

Q33: An application with a 99.95% uptime requirement is down for 30 minutes monthly. Does it meet the SLA?

☒ **C) No, it exceeds the allowed downtime**

Q34: A system processes 5,000 transactions per second and runs continuously. How many transactions are processed per day?

☒ **D) 432 million transactions**

8. Spiral Model & Risk Management

Q35: A Spiral Model project has a budget of \$250,000 and allocates 40% to risk assessment. How much is allocated?

☒ **B) \$100,000**

Q36: If an unexpected risk causes a 15% budget overrun, what is the new project cost?

☒ **D) \$287,500**

9. Software Development & Defect Management

Q37: A 250,000 LOC project has a defect density of 0.45 defects/KLOC. How many total defects are expected?

☒ **C) 112 defects**

Q38: After fixing 85% of defects, how many defects remain?

☒ **B) 17 defects**

10. Risk Management

Q39: A project has a 40% probability of scope changes, with a high impact factor of 3. What is the risk exposure?

☒ **C) 1.2**

Q40: Which risk strategy is best for mitigating scope creep in an agile project?

☒ **B) Continuous stakeholder feedback**

1. Feasibility Study in Software Engineering

Q41: What is the total initial investment required for a software project with the following costs?

- Development Costs: \$120,000
- Hardware Costs: \$40,000
- Training Costs: \$30,000

A) \$160,000

B) \$190,000

C) \$170,000

D) \$150,000

☒ **Answer: B) \$190,000**

2. Net Present Value (NPV) Calculation

Q42: If the Net Cash Flow per year is \$60,000 and the discount rate is 10%, what is the present value for year 3?

A) \$49,587

B) \$45,087

C) \$48,760

D) \$50,200

✓ **Answer: A) \$49,587**

3. COCOMO Model Calculation

Q43: In the COCOMO II model, what is the effort (in person-months) required for a project with 1000 KLOC in the organic mode, given that the effort equation is **Effort** = **2.94** × (Size)^{1.05}?

- A) 2,980 person-months
- B) 3,243 person-months**
- C) 3,500 person-months
- D) 3,200 person-months

✓ **Answer: B) 3,243 person-months**

4. Cyclomatic Complexity

Q44: What is the cyclomatic complexity (CC) of a module with 10 edges and 7 nodes?

- A) 5
- B) 6**
- C) 4
- D) 3

✓ **Answer: B) 6** (Formula: $CC = E - N + 2$)

5. Defect Density Calculation

Q45: A project has 200 defects in 100,000 lines of code. What is its defect density per KLOC?

- A) 1.5
- B) 2.0**
- C) 2.5
- D) 3.0

✓ **Answer: B) 2.0** (Defect Density = $200 / 100 = 2.0$ defects per KLOC)

6. Risk Management in Software Development

Q46: If a project has a 40% chance of a major bug occurring, with a resolution cost of \$15,000, what is the expected cost of the bug?

- A) \$4,500
- B) \$6,000**
- C) \$5,000
- D) \$6,500

✓ **Answer: B) \$6,000** (Expected Cost = $0.40 \times \$15,000 = \$6,000$)

7. Rational Unified Process (RUP)

Q47: If the Elaboration phase of a project costs \$3,000 per week for 8 weeks, what is the total cost for this phase?

- A) \$18,000
- B) \$20,000
- C) \$24,000
- D) \$30,000

☒ **Answer: C) \$24,000**

8. Transaction Rate Calculation

Q48: If a system processes 600 transactions per hour, how many transactions occur per second?

- A) 0.17
- B) 0.25
- C) 0.5
- D) 0.167

☒ **Answer: D) 0.167 transactions/second**

9. Throughput in Microservices

Q49: If a system has 50 concurrent users and each makes 2 requests per second, what is the system's total request rate?

- A) 50 requests/sec
- B) 100 requests/sec
- C) 75 requests/sec
- D) 150 requests/sec

☒ **Answer: B) 100 requests/sec**

10. Equivalence Partitioning

Q50: How many equivalence classes exist for an input range of 1 to 500?

- A) 2
- B) 3
- C) 4
- D) 5

☒ **Answer: B) 3 (Valid range, <1, >500)**

11. Git Repository Analysis

Q51: If a team of 10 developers makes 5 commits per day for 30 days, how many total commits are made?

- A) 1,200
- B) 1,500
- C) 1,400
- D) 1,600

☒ **Answer: B) 1,500**

12. Code Review Effort

Q52: If each code review takes 10 minutes and there are 300 commits, how many hours are spent on reviews?

- A) 40
- B) 50
- C) 60
- D) 55

☒ **Answer: C) 50 hours**

13. Merge Conflict Resolution

Q53: If 200 merge conflicts take 15 minutes each, how many total hours are spent?

- A) 30
- B) 40
- C) 50
- D) 45

☒ **Answer: B) 50 hours**

14. Productivity Loss

Q54: If a team loses 10% of 500 working hours due to merge conflicts, how many hours are lost?

- A) 40
- B) 50
- C) 60
- D) 70

☒ **Answer: B) 50 hours**

15. Functional Requirements

Q55: Which of the following is a functional requirement?

- A) Response time < 2 sec
- B) UI must be intuitive
- C) The system must process 1,000 orders per hour
- D) The system must be secure

☒ **Answer: C) The system must process 1,000 orders per hour**

16. Non-Functional Requirements

Q56: What is an example of a non-functional requirement?

- A) Users should be able to log in
- B) The system must support 500 concurrent users
- C) The checkout process should include tax calculation
- D) Admins should be able to manage users

☒ **Answer: B) The system must support 500 concurrent users**

17. Software Maintenance Types

Q57: Fixing a critical bug in deployed software is an example of:

- A) Adaptive Maintenance
- B) Corrective Maintenance
- C) Perfective Maintenance
- D) Preventive Maintenance

☒ **Answer: B) Corrective Maintenance**

18. Software Development Models

Q58: Which model is best for projects with unclear requirements?

- A) Waterfall
- B) Spiral
- C) V-Model
- D) Big Bang

☒ **Answer: B) Spiral**

19. Project Estimation

Q59: The most common size estimation metric in software engineering is:

- A) Function Points
- B) KLOC
- C) Use Case Points
- D) Story Points

✓ **Answer: B) KLOC**

20. Software Testing

Q60: Which testing type is performed without knowing internal code structure?

- A) Unit Testing
- B) Integration Testing
- C) Black-Box Testing
- D) White-Box Testing

✓ **Answer: C) Black-Box Testing**

1. Agile Methodology & Principles

61. **A software development team follows Agile practices and frequently delivers working software. However, stakeholders demand extensive documentation. What should the team prioritize?**

- a) Reduce working software deliveries to create detailed documents
- b) Balance documentation while still prioritizing working software
- c) Ignore documentation entirely
- d) Deliver only documents and postpone software release

Answer: b) Balance documentation while still prioritizing working software

62. **An Agile team receives frequent change requests from customers. According to Agile principles, how should they respond?**

- a) Accept all changes at any stage of development
- b) Reject changes after the initial planning phase
- c) Welcome changes even late in development
- d) Ask customers to wait until the next release

Answer: c) Welcome changes even late in development

2. Software Design & Architecture

63. **An e-commerce website is experiencing slow response times during peak hours. What architectural change should be considered?**

- a) Move to a microservices architecture
- b) Convert it to a monolithic architecture
- c) Use a single database instance for all transactions
- d) Remove caching mechanisms to free up memory

Answer: a) Move to a microservices architecture

64. **Which architectural pattern is best suited for real-time stock trading applications?**

- a) Monolithic architecture
- b) Client-server model
- c) Event-driven architecture
- d) Layered architecture

Answer: c) Event-driven architecture

3. Software Testing & Quality Assurance

65. **A banking application requires strict security testing to protect financial transactions. Which testing type is most critical?**

- a) Usability Testing
- b) Performance Testing
- c) Penetration Testing
- d) A/B Testing

Answer: c) Penetration Testing

66. **A software engineer wants to ensure that all code paths are executed during testing. What technique should they use?**

- a) Black-box testing
- b) Equivalence partitioning
- c) Statement and branch coverage testing
- d) Beta testing

Answer: c) Statement and branch coverage testing

4. Software Estimation & Cost Analysis

67. **A software project requires 80 person-months of effort. Using the COCOMO model with $B = 2.8$ and $E = 1.2$, what happens if the complexity increases, and B changes to 3.2?**

- a) Effort remains unchanged
- b) Effort increases due to higher complexity
- c) Effort decreases as the team becomes more efficient
- d) Effort reduces, but the project duration remains the same

Answer: b) Effort increases due to higher complexity

68. **In the Rational Unified Process (RUP), which phase requires the highest effort in a large-scale project?**

- a) Inception
- b) Elaboration
- c) Construction
- d) Transition

Answer: c) Construction

5. Risk Management in SDLC

69. **Which risk is highest in an Agile development model?**

- a) Changing requirements
- b) High documentation overhead
- c) Fixed scope and timeline
- d) Lack of collaboration

Answer: a) Changing requirements

70. **Which strategy is most appropriate for handling unpredictable third-party service failures in a cloud-based application?**

- a) Implement circuit breakers and fallback mechanisms
- b) Remove dependency on third-party services
- c) Manually restart failed services
- d) Avoid redundancy to simplify maintenance

Answer: a) Implement circuit breakers and fallback mechanisms

6. Software Security & Safety

71. **A nuclear power plant control system requires high reliability and fault tolerance. Which standard should the software adhere to?**

- a) ISO 26262
- b) IEC 61513
- c) GDPR
- d) Agile Manifesto

Answer: b) IEC 61513

72. **An online banking system needs to prevent unauthorized access. What is the best approach?**

- a) Implement multi-factor authentication
- b) Allow single sign-on without additional verification
- c) Store passwords in plaintext for easy access
- d) Disable security logs to improve performance

Answer: a) Implement multi-factor authentication

7. Software Maintenance & Evolution

73. **A legacy system needs frequent updates, but maintenance is difficult due to tight coupling. What should be done?**

- a) Continue maintaining as is
- b) Refactor the code into smaller, modular components
- c) Rewrite the entire application from scratch
- d) Ignore updates to avoid further issues

Answer: b) Refactor the code into smaller, modular components

74. **Which software maintenance type is involved when fixing security vulnerabilities in an application?**

- a) Corrective maintenance
- b) Adaptive maintenance
- c) Perfective maintenance
- d) Preventive maintenance

Answer: a) Corrective maintenance

8. Software Project Management

75. **A project team follows the Waterfall model but struggles with delayed feedback. What is the best solution?**

- a) Switch to Agile for faster feedback loops
- b) Continue using Waterfall and extend the timeline
- c) Reduce testing time to speed up development
- d) Remove customer involvement to avoid interruptions

Answer: a) Switch to Agile for faster feedback loops

76. **A company is considering outsourcing software development. What is the biggest risk?**

- a) Higher development cost
- b) Intellectual property theft and data security issues
- c) Increased software quality
- d) Faster time to market

Answer: b) Intellectual property theft and data security issues

9. Software Reliability & Performance

77. **A medical software system must ensure 100% uptime. What architecture should be implemented?**

- a) Distributed fault-tolerant architecture
- b) Single-server architecture
- c) Waterfall model with documentation
- d) Minimalistic design without redundancy

Answer: a) Distributed fault-tolerant architecture

78. **A mobile application must run smoothly on both low-end and high-end devices. What should be the primary focus?**

- a) High-end device optimization only
- b) Minimizing resource usage and efficient memory management
- c) Using high-resolution assets for better visuals
- d) Disabling features to simplify development

Answer: b) Minimizing resource usage and efficient memory management

10. Software Testing Strategies

79. **A web-based shopping platform has performance issues with high user traffic. What testing method should be used?**

- a) Load testing
- b) Unit testing
- c) Alpha testing
- d) Regression testing

Answer: a) Load testing

80. **A developer wants to test how well the system recovers from crashes. What testing type is appropriate?**

- a) Functional testing
- b) Recovery testing
- c) Integration testing
- d) Static testing

Answer: b) Recovery testing

Scrum Master Responsibilities

81. Which of the following is NOT a primary responsibility of a Scrum Master? a)

Facilitating Scrum events

b) Assigning tasks to developers

c) Removing impediments

d) Coaching the team

Answer: b) Assigning tasks to developers

Historical Data in Cost Estimation

82. How can historical project data improve cost estimation accuracy?

a) By increasing project complexity

b) By eliminating the need for expert judgment

- c) By providing baselines for effort, time, and costs
- d) By ensuring that estimates never change

Answer: c) By providing baselines for effort, time, and costs

Software Design Quality

83. Which characteristic of software design ensures it can handle increased workload efficiently?

- a) Maintainability
- b) Scalability
- c) Usability
- d) Security

Answer: b) Scalability

Boundary Value Analysis

84. Which of the following is a valid boundary value test case for a date input system ($1 \leq \text{day} \leq 31, 1 \leq \text{month} \leq 12$)?

- a) (0, 1, 1995)
- b) (32, 5, 2000)
- c) (1, 1, 1990)
- d) (15, 6, 2005)

Answer: c) (1, 1, 1990)

Software Estimation

85. A project consists of 50,000 LOC, with a developer productivity of 25 LOC/hour. How many person-hours are required?

- a) 2000
- b) 3000
- c) 5000
- d) 1250

Answer: c) 2000

CASE Tool Benefits

86. What is a key advantage of using CASE tools in software development?

- a) Increases manual effort
- b) Reduces consistency in development
- c) Enhances automation in design and coding
- d) Slows down the software development process

Answer: c) Enhances automation in design and coding

Class Diagrams

87.Which UML relationship best represents the "Books are associated with Authors" scenario in a library management system?

- a) One-to-One
- b) Many-to-Many
- c) One-to-Many
- d) None

Answer: b) Many-to-Many

SafeHome System UI

88.Which menu item in a home security system is critical for handling emergency situations?

- a) Settings
- b) Help & Support
- c) Security Management
- d) Camera View

Answer: c) Security Management

Cyclomatic Complexity

89.A function has three independent decision points. What is its cyclomatic complexity?

- a) 2
- b) 3
- c) 4
- d) 5

Answer: d) 5 (Formula: $CC = \text{Number of decision points} + 1$)

White-Box Testing

90.Which test case is necessary for branch coverage in a sorting algorithm?

- a) Already sorted array
- b) Reverse sorted array
- c) Array with duplicate elements
- d) All of the above

Answer: d) All of the above

Change Management Challenges

91.Which is a major challenge in change management?

- a) Increased project control
- b) Enhanced quality
- c) Resistance to change
- d) Improved transparency

Answer: c) Resistance to change

Risk Mitigation

92.What is the primary goal of a risk mitigation strategy?

- a) To ignore low-probability risks
- b) To eliminate all project risks
- c) To reduce the impact of risks on project success
- d) To increase project complexity

Answer: c) To reduce the impact of risks on project success

Software Maintenance

93.Which type of software maintenance focuses on modifying the system to adapt to environmental changes?

- a) Corrective
- b) Adaptive
- c) Perfective
- d) Preventive

Answer: b) Adaptive

Earned Value Analysis

94.If a project has an Earned Value (EV) of \$10,000 and an Actual Cost (AC) of \$12,000, what is the Cost Performance Index (CPI)?

- a) 1.2
- b) 0.83
- c) 1.5
- d) 1.0

Answer: b) 0.83 (CPI = EV/AC)

Agile vs. Waterfall

95.Which statement is TRUE regarding Agile development compared to the Waterfall model?

- a) Agile follows a strict sequential process
- b) Waterfall allows frequent iteration and feedback
- c) Agile is more adaptive to changing requirements
- d) Waterfall is better for rapidly evolving projects

Answer: c) Agile is more adaptive to changing requirements

Functional vs. Non-Functional Requirements

96.Which of the following is an example of a non-functional requirement?

- a) The system shall allow users to log in

- b) The application shall process 100 transactions per second
- c) The user shall be able to reset their password
- d) The system shall generate monthly reports

Answer: b) The application shall process 100 transactions per second

Software Metrics

97. Which metric is used to measure software complexity?

- a) Function Points
- b) Cyclomatic Complexity
- c) Code Coverage
- d) Lines of Code

Answer: b) Cyclomatic Complexity

Defect Prevention Strategies

98. Which strategy is most effective in preventing software defects?

- a) Performing extensive manual testing
- b) Eliminating all documentation
- c) Implementing code reviews and static analysis
- d) Relying solely on end-user testing

Answer: c) Implementing code reviews and static analysis

Software Deployment Risks

99. What is a potential risk when deploying software without rollback mechanisms?

- a) Improved system stability
- b) Difficulty in reverting to a previous version
- c) Faster feature release
- d) Decreased testing effort

Answer: b) Difficulty in reverting to a previous version

Software Process Models

100. Which software process model is best suited for projects with unclear and evolving requirements?

- a) Waterfall
- b) V-Model
- c) Agile
- d) Spiral

Answer: c) Agile

Section 1: Agile Methodology & Scrum (40 Questions)

101. A software team using Agile methodology is working in 2-week sprints. If the team has planned 8 sprints, what is the total duration of the project in weeks?
- a) 12 weeks
 - b) 16 weeks
 - c) 20 weeks
 - d) 24 weeks
 - **Answer: b) 16 weeks**
102. Which Agile principle emphasizes continuous feedback from customers?
- a) Working software is the primary measure of progress
 - b) Deliver working software frequently
 - c) Customer collaboration over contract negotiation
 - d) Build projects around motivated individuals
 - **Answer: c) Customer collaboration over contract negotiation**
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Section 2: Rational Unified Process (RUP) (30 Questions)

103. A project follows RUP and is in the elaboration phase. The planned duration is 3 months, and 40% of work is completed. How much time remains in this phase?
- a) 1.2 months
 - b) 1.8 months
 - c) 2 months
 - d) 0.8 months
 - **Answer: b) 1.8 months**
104. Which RUP phase focuses on risk mitigation and refining system architecture?
- a) Inception
 - b) Elaboration
 - c) Construction
 - d) Transition
 - **Answer: b) Elaboration**
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Section 3: COCOMO Model & Software Estimation (40 Questions)

105. A software project requires 50,000 LOC and follows a semi-detached COCOMO model. Using parameters $a=3.0$ and $b=1.12$, what is the estimated effort (in person-months)?
- a) 150
 - b) 189.29
 - c) 200
 - d) 220
 - **Answer: b) 189.29**
106. What happens when analyst capability increases in the COCOMO model?
- a) Development effort increases

- b) Development effort decreases
 - c) No impact
 - d) Project duration extends
 - **Answer:** b) Development effort decreases
-

Section 4: Software Requirement Engineering (40 Questions)

107. **A project starts with 120 requirements. After elicitation (15% increase), analysis (10% refinement), and validation (5% modification), how many final requirements exist?**
- a) 132
 - b) 138
 - c) 160
 - d) 152
 - **Answer:** d) 152
108. **Which prioritization technique is used to categorize features into Must Have, Should Have, Could Have, and Won't Have?**
- a) MoSCoW Method
 - b) SWOT Analysis
 - c) PERT Estimation
 - d) Cost-Benefit Analysis
 - **Answer:** a) MoSCoW Method
-

Section 5: Software Testing & Quality Assurance (40 Questions)

109. **Unit testing requires 80 hours. If integration testing takes 40% of unit testing, how many hours are required for integration testing?**
- a) 20 hours
 - b) 32 hours
 - c) 40 hours
 - d) 50 hours
 - **Answer:** c) 40 hours
110. **Which testing type ensures different software modules work together correctly?**
- a) Unit Testing
 - b) Integration Testing
 - c) Regression Testing
 - d) Load Testing
 - **Answer:** b) Integration Testing
-

Section 6: Software Design Patterns (30 Questions)

111. Which design pattern ensures that only one instance of a class is created?

- a) Factory Pattern
- b) Singleton Pattern
- c) Observer Pattern
- d) Prototype Pattern
- **Answer:** b) Singleton Pattern

112. What is the primary purpose of the Observer pattern?

- a) Allow multiple objects to be notified of state changes
 - b) Manage object instantiation
 - c) Ensure data security
 - d) Reduce redundancy
 - **Answer:** a) Allow multiple objects to be notified of state changes
-

Section 7: Software Architecture & Design (30 Questions)

113. In a distributed system handling millions of users, which component is responsible for distributing incoming network traffic?

- a) Load Balancer
- b) Microservices
- c) API Gateway
- d) Database Replication
- **Answer:** a) Load Balancer

114. If an e-commerce system has microservices handling orders, payments, and user authentication separately, what architectural style is being used?

- a) Monolithic Architecture
 - b) Layered Architecture
 - c) Microservices Architecture
 - d) Client-Server Architecture
 - **Answer:** c) Microservices Architecture
-

Section 8: Case Study-Based Questions (30 Questions)

115. A CRM system project has an initial cost of \$170,000. If the net annual benefit is \$80,000 and the maintenance cost is \$30,000, what is the annual net cash flow?

- a) \$40,000
- b) \$50,000
- c) \$60,000

- d) \$80,000
- **Answer:** b) \$50,000

116. **An online banking system requires a page load time of 3 seconds. If the measured response time is 2.8 seconds, what is the compliance rate?**

- a) 95%
- b) 100%
- c) 107.14%
- d) 120%
- **Answer:** c) 107.14%

1. Agile Methodology & Scrum (50 Questions)

Basic Concepts

117. **Which of the following is NOT a core value of Agile methodology?**
- a) Individuals and interactions over processes and tools
 - b) Working software over comprehensive documentation
 - c) Strict adherence to initial project plans
 - d) Customer collaboration over contract negotiation
 - **Answer:** c) Strict adherence to initial project plans
118. **In Agile, a sprint typically lasts for:**
- a) 1 week
 - b) 2 weeks
 - c) 4 weeks
 - d) 6 weeks
 - **Answer:** b) 2 weeks
119. **What is the primary goal of a Sprint Retrospective?**
- a) Review the product backlog
 - b) Discuss team performance and identify improvements
 - c) Plan the next sprint
 - d) Update documentation
 - **Answer:** b) Discuss team performance and identify improvements
120. **What is a Scrum Master responsible for?**
- a) Assigning tasks to developers
 - b) Defining technical architecture
 - c) Facilitating Scrum processes and removing impediments
 - d) Writing user stories
 - **Answer:** c) Facilitating Scrum processes and removing impediments

2. Rational Unified Process (RUP) (50 Questions)

Project Planning & Execution

121. Which phase of RUP focuses on risk mitigation and refining system architecture?
- a) Inception
 - b) Elaboration
 - c) Construction
 - d) Transition
 - **Answer: b) Elaboration**
122. If a project follows the RUP methodology and the Elaboration phase is planned for 3 months, but 40% of work is completed, how much time remains for this phase?
- a) 1.2 months
 - b) 1.8 months
 - c) 2 months
 - d) 0.8 months
 - **Answer: b) 1.8 months**
123. The total duration of a project using RUP is 12 months. If the elapsed time so far is 2.2 months, what percentage of the total project duration has passed?
- a) 10%
 - b) 15%
 - c) 18.33%
 - d) 20%
 - **Answer: c) 18.33%**
-

3. COCOMO Model & Estimation (50 Questions)

Effort Estimation & Cost Analysis

124. Which of the following is a primary objective of the COCOMO model?
- a) Project documentation
 - b) Software effort and cost estimation
 - c) Code optimization
 - d) Agile sprint management
 - **Answer: b) Software effort and cost estimation**
125. A software project is estimated to require 50,000 lines of code (LOC) and follows a semi-detached COCOMO model. Using parameters $a=3.0$, $b=1.12$, what is the estimated effort (in person-months)?
- a) 100
 - b) 150
 - c) 189.29
 - d) 200
 - **Answer: c) 189.29**
126. What is the impact of increasing analyst capability in the COCOMO model?

- a) Increases development effort
 - b) Decreases development effort
 - c) Has no impact
 - d) Extends the development timeline
 - **Answer:** b) Decreases development effort
-

4. Software Requirement Engineering (50 Questions)

Requirement Analysis & Validation

125. If a project starts with 150 requirements and 20% additional emerge in the elicitation phase, how many requirements are there after this phase?
- a) 120
 - b) 150
 - c) 180
 - d) 170
 - **Answer:** d) 170
126. Which of the following is NOT a category in the MoSCoW prioritization method?
- a) Must Have
 - b) Should Have
 - c) Could Have
 - d) Won't Fix
 - **Answer:** d) Won't Fix
127. What is the main objective of the validation phase in software development?
- a) Remove unnecessary requirements
 - b) Ensure requirements meet user needs
 - c) Implement requirements
 - d) Finalize cost estimation
 - **Answer:** b) Ensure requirements meet user needs
-

5. Software Testing & Quality Assurance (50 Questions)

Unit, Integration & System Testing

128. Which type of testing is performed to validate that different modules work together?
- a) Unit Testing

- b) Integration Testing
- c) Regression Testing
- d) Load Testing
- **Answer:** b) Integration Testing

129. **If unit testing requires 80 hours, and integration testing takes 40% of that time, how many hours are required for integration testing?**

- a) 20 hours
- b) 32 hours
- c) 40 hours
- d) 50 hours
- **Answer:** c) 40 hours

130. **The primary purpose of system testing is to:**

- a) Test individual modules
- b) Evaluate system performance
- c) Ensure software meets end-user requirements
- d) Optimize code efficiency
- **Answer:** c) Ensure software meets end-user requirements

6. Software Design Patterns (50 Questions)

Design Principles & Architecture

131. **Which design pattern ensures that only one instance of a class is created?**

- a) Factory Pattern
- b) Singleton Pattern
- c) Observer Pattern
- d) Prototype Pattern
- **Answer:** b) Singleton Pattern

132. **Which of the following design patterns falls under the structural category?**

- a) Strategy
- b) Adapter
- c) Observer
- d) Factory
- **Answer:** b) Adapter

133. **What is the main purpose of the Observer pattern?**

- a) Allow multiple objects to be notified when a change occurs

- b) Manage object instantiation
- c) Ensure data security
- d) Reduce code redundancy
- **Answer:** a) Allow multiple objects to be notified when a change occurs

134. **A software project is estimated to take 12 months with a team of 5 developers. If the team size is increased to 10 developers, what is the new estimated duration assuming linear scalability?**

- A) 6 months
- B) 8 months
- C) 10 months
- D) 12 months

Answer: A) 6 months

135. **If a project has a defect density of 0.5 defects per 100 lines of code and the project has 10,000 lines of code, how many defects are expected?**

- A) 25
- B) 50
- C) 75
- D) 100

Answer: B) 50

136. **A project manager estimates that 20% of the project budget will be spent on testing. If the total budget is \$200,000, how much will be allocated for testing?**

- A) \$20,000
- B) \$30,000
- C) \$40,000
- D) \$50,000

Answer: C) \$40,000

137. **In a project, if the planned value (PV) is \$100,000, the earned value (EV) is \$80,000, and the actual cost (AC) is \$90,000, what is the cost performance index (CPI)?**

- A) 0.89
- B) 0.95
- C) 1.11
- D) 1.25

Answer: A) 0.89

138. **If a software application has a response time of 2 seconds under normal load and 5 seconds under peak load, what is the percentage increase in response time during peak load?**

- A) 100%
- B) 150%
- C) 200%
- D) 250%

Answer: B) 150%

139. A team completes 60% of a project in 4 months. If the remaining work is expected to take 3 months, what is the total estimated duration of the project?

- A) 6 months
- B) 7 months
- C) 8 months
- D) 9 months

Answer: C) 8 months

140. If a project has a risk probability of 30% and a potential impact of \$50,000, what is the expected monetary value (EMV) of the risk?

- A) \$10,000
- B) \$15,000
- C) \$20,000
- D) \$25,000

Answer: A) \$15,000

141. A software system is expected to handle 1,000 transactions per minute. If the system can currently handle 800 transactions per minute, what is the percentage shortfall in capacity?

- A) 10%
- B) 20%
- C) 25%
- D) 30%

Answer: B) 20%

142. If a project is 75% complete and has a budget of \$200,000, what is the earned value (EV) of the project?

- A) \$150,000
- B) \$175,000
- C) \$200,000
- D) \$225,000

Answer: A) \$150,000

143. A software application has a failure rate of 0.02 failures per hour. If it runs for 1,000 hours, how many failures are expected?

- A) 10
- B) 20
- C) 30
- D) 40

Answer: B) 20

144. If a project has a schedule variance (SV) of -\$5,000 and an earned value (EV) of \$50,000, what is the planned value (PV)?

- A) \$45,000
- B) \$55,000
- C) \$60,000
- D) \$65,000

Answer: B) \$55,000

145. A software development team has a velocity of 30 story points per sprint. If the team has 5 sprints to complete a project with 150 story points, will they complete the project on time?

- A) Yes
- B) No

Answer: A) Yes

146. If a project has a budget of \$500,000 and is currently 40% complete, what is the budget at completion (BAC)?
- A) \$200,000
 - B) \$300,000
 - C) \$400,000
 - D) \$500,000
- Answer: D) \$500,000
147. A software system is designed to process 200 requests per second. If it currently processes 150 requests per second, what is the percentage of utilization?
- A) 60%
 - B) 70%
 - C) 75%
 - D) 80%
- Answer: C) 75%
148. If a project has a cost variance (CV) of \$10,000 and an actual cost (AC) of \$90,000, what is the earned value (EV)?
- A) \$80,000
 - B) \$90,000
 - C) \$100,000
 - D) \$110,000
- Answer: C) \$100,000
149. A project is expected to take 8 months with a budget of \$400,000. If the project is completed in 6 months, what is the cost performance index (CPI) if the actual cost is \$350,000?
- A) 1.14
 - B) 1.20
 - C) 1.25
 - D) 1.30
- Answer: A) 1.14
150. If a software application has a mean time to failure (MTTF) of 1,000 hours, what is the expected number of failures in a 10,000-hour period?
- A) 5
 - B) 10
 - C) 15
 - D) 20
- Answer: B) 10
151. A project has a planned value (PV) of \$200,000, an earned value (EV) of \$180,000, and an actual cost (AC) of \$210,000. What is the schedule performance index (SPI)?
- A) 0.85
 - B) 0.90
 - C) 1.00
 - D) 1.10
- Answer: A) 0.90
152. If a project has a total of 1,000 tasks and 600 tasks are completed, what is the percentage of completion?
- A) 50%
 - B) 60%
 - C) 70%

- D) 80%

Answer: B) 60%

153. A software system is expected to have a 99.9% uptime. How many hours of downtime can be tolerated in a year?

- A) 8.76 hours
- B) 17.52 hours
- C) 43.8 hours
- D) 87.6 hours

Answer: A) 8.76 hours

154. If a project has a budget of \$1,000,000 and is currently 50% complete, what is the earned value (EV) of the project?

- A) \$400,000
- B) \$500,000
- C) \$600,000
- D) \$700,000

Answer: B) \$500,000

155. A software application has a defect rate of 0.1 defects per function point. If the application has 200 function points, how many defects are expected?

- A) 10
- B) 15
- C) 20
- D) 25

Answer: C) 20

156. If a project has a cost performance index (CPI) of 1.2 and an actual cost (AC) of \$50,000, what is the earned value (EV)?

- A) \$40,000
- B) \$60,000
- C) \$70,000
- D) \$80,000

Answer: B) \$60,000

157. A project is expected to take 10 months with a budget of \$300,000. If the project is completed in 8 months, what is the cost performance index (CPI) if the actual cost is \$250,000?

- A) 1.20
- B) 1.25
- C) 1.30
- D) 1.40

Answer: B) 1.20

158. If a project has a schedule variance (SV) of \$15,000 and a planned value (PV) of \$100,000, what is the earned value (EV)?

- A) \$85,000
- B) \$95,000
- C) \$115,000
- D) \$125,000

Answer: B) \$115,000

159. A software application has a response time of 1 second under normal load and 3 seconds under peak load. What is the percentage increase in response time during peak load?

- A) 100%
- B) 200%

- C) 300%
- D) 400%

Answer: B) 200%

160. **If a project has a budget of \$500,000 and is currently 30% complete, what is the earned value (EV) of the project?**

- A) \$100,000
- B) \$150,000
- C) \$200,000
- D) \$250,000

Answer: C) \$150,000

161. **A software system is designed to handle 1,000 users concurrently. If it currently supports 800 users, what is the percentage of utilization?**

- A) 70%
- B) 75%
- C) 80%
- D) 85%

Answer: B) 80%

162. **If a project has a cost variance (CV) of -\$20,000 and an earned value (EV) of \$100,000, what is the actual cost (AC)?**

- A) \$80,000
- B) \$90,000
- C) \$100,000
- D) \$120,000

Answer: D) \$120,000

163. **A software application has a defect density of 0.05 defects per line of code. If the application has 50,000 lines of code, how many defects are expected?**

- A) 1,000
- B) 2,500
- C) 3,000
- D) 4,000

Answer: A) 2,500